

Erdős Problem #412

EP-412

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $\sigma_1(n) = \sigma(n)$, the sum of divisors function, and $\sigma_k(n) = \sigma(\sigma_{k-1}(n))$.

Is it true that, for every $m, n \geq 2$, there exist some i, j such that $\sigma_i(m) = \sigma_j(n)$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>